



11/6/18

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ESE

Class: - TE EXTC

Sem: - VI

Total Marks: - 100

Duration: - 3 Hour

Subject: - Operating System (ETC605)

Year: - 2017-18

- 1) (a) List various software components of a typical operating system and hence enumerate various functions and services provided by an operating system. [CO1] [10]

Various Components: Kernel, Device Drivers, System Programs

Services:- User Interface, Program execution, I/O operation, File-system manipulation, Communications, Error detection, Resource allocation, Accounting, Protection and security

- (b) What is a semaphore? Write a small section of code for producer process and consumer process with bounded buffer problem using semaphores to eliminate critical section problem. [CO2] [10]

Definition of a semaphore, wait() and signal() operations

Mutex=1, empty=n, full=0

Producer process:-

```
Do{  
  
    Wait(empty);  
  
    Wait(mutex);  
  
    ...  
  
    ...  
  
    Signal(mutex);  
  
    Signal(full);  
  
} while(true);
```

Consumer process:-

```
Do{  
  
    Wait(full);
```

```
Wait(mutex);
```

```
...
```

```
...
```

```
Signal(mutex);
```

```
Signal(empty);
```

```
}while(true);
```

- 2) (a) Consider a following set of processes, with length of CPU bursts given in milliseconds as follows:

Process	Burst Time	Arrival Time
P1	10	0
P2	29	0
P3	3	0
P4	7	0
P5	12	0

Draw the Gantt charts for FCFS, Non Preemptive SJF and RR (Quantum=10)

What is the waiting time of each process for each of the above algorithms?

What is the turnaround time of each process for each of the above algorithms?

Which algorithm results in the minimum average waiting time? [CO2]

[10]

OR

Consider a following set of processes, with length of CPU bursts given in milliseconds as follows:

Process	Burst Time	Arrival Time
P1	14	0
P2	8	0
P3	21	0
P4	10	0

Draw the Gantt charts for FCFS, Non Preemptive SJF and RR (Quantum=10)

What is the waiting time of each process for each of the above algorithms?

What is the turnaround time of each process for each of the above algorithms?

Which algorithm results in the minimum average waiting time? [CO2]

[10]

- 2) (b) With respect to UNIX OS, Draw and discuss the process state transition diagram.

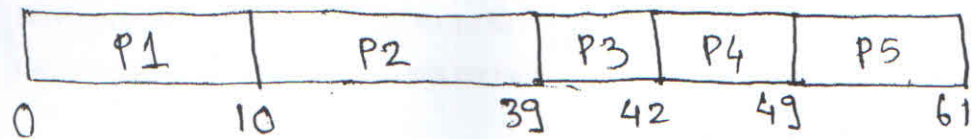
[CO3]

[10]

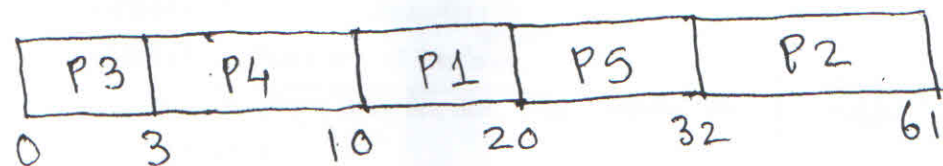
Q.2 @ (First Part)

Gantt charts

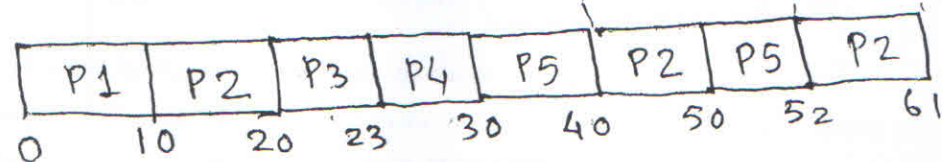
① FCFS:-



② Non Pre emptive SJF:-



③ RR (Quantum = 10):-



Process	Burst Time	Arrival Time	Priority	FCFS		Non Pre-SJF		---		RR (Q=10)	
				TW	TA	TW	TA	---	---	TW	TA
P1	10	0	---	0	10	10	20	---	---	0	10
P2	29	0	---	10	39	32	61	---	---	32	61
P3	3	0	---	39	42	0	3	---	---	20	23
P4	7	0	---	42	49	3	10	---	---	23	30
P5	12	0	---	49	61	20	32	---	---	40	52
Average value =				28		13		---	---	23	

Five process states: New, Ready, Running, Blocked, Halted
Draw the process state transition diagram with justification.

OR

Discuss various security threats in UNIX OS and also enumerate the features included to address these threats. [CO3] [10]

User Accounts

File Permissions

Data Verification

Encrypted Storage

Secure Remote Access with OpenSSH

Software Management

Host Integrity Testing

System Recovery

Resource Allocation Controls

Monitoring and Audit Facilities

The System Firewall

Application Isolation

A Note on Viruses and Malware

- 3) (a) Discuss the mechanism of translating the virtual address (logical address) to its respective physical address for a particular operating system. Consider the concept of segmentation and the case study of Linux OS while writing your answer. [CO3] [10]

Memory is divided into segments. Then a segment can be divided into pages of fixed size.

Hence, first describe the concept of segmentation with virtual address definition. [04]

Second, describe the segment translation process with diagram. [06]

OR

With respect to Linux OS, discuss the file management in general and file allocation method used in particular. (inode structure). [CO3] [10]

LINUX inode structure has 15 block pointers. First 12 are direct data block pointers. 13th slot is known as single indirect pointer. 14th slot is known as double indirect pointer. 15th slot is known as triple indirect pointer. The complete detailed diagram of the inode structure is to be drawn.

- 3) (b) Consider the following page reference string:

7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1

How many page faults would occur for the following replacement algorithms, assuming three page frames? Remember that all frames are initially empty, so your first unique pages will cost one page each.

FIFO

LRU

OPTIMAL [CO2]

[10]

- 4) (a) With respect to file systems, discuss (a) File Attributes (b) File Operations (c) Directory Structure (Single level, Two level and Tree level) [CO2] [10]

File Attributes:- Name, Identifier, Size, Type, Location, Protection, Time-Date-User-Identifier [3]

File Operations:- Create, Write, Read, Reposition, Delete, Truncate [3]

Draw diagrams for Single, Two and Tree level directory structure. [4]

- 4) (b) Assume the number of tracks are 200 numbered from 0 to 199 on a particular hard disk. Assume the current head position is on the track number 53. Consider a disk queue with requests for I/O to blocks on cylinders as

98, 183, 37, 122, 14, 124, 65, 67

Calculate the average seek time in terms of number of cylinders for the four different disk arm scheduling algorithms (a) FCFS (b) SSTF (c) SCAN (d) C-SCAN [CO2] [10]

- 5) (a) Compare Static priority scheduling algorithm vs Dynamic priority algorithms. [CO3] [10]

- 5) (b) Consider a system with three tasks, which we'll call Task 1, Task 2 and Task 3. Assume all are periodic tasks with periods T₁, T₂ and T₃. Each has a deadline that is the beginning of its next cycle.

Task 1 has P₁ = 4 ms, C₁ = 1 ms, Task 2 has P₂ = 10 ms, C₂ = 2 ms and Task 3 has P₃ = 15 ms, C₃ = 4 ms. Consider static priority scheduling (RMA), draw the time space diagram and calculate processor utilization, schedulability. [CO3] [10]

Ans 5(a):

One major limitation of fixed-priority scheduling is that it is not always possible to fully utilize the CPU.

Static (RMA) only guarantees feasibility at 69% utilization, EDF guarantees it at 100%

Dynamic (EDF) is complicated enough to have unacceptable overhead

Dynamic More complicated than RMA: harder to analyze, Less predictable: can't guarantee which process runs when

Dynamic algorithm behaves unpredictably at overloads, and suffers more from overloads than Static (RMA)

Ans 5(b):

U (schedule bound) = 0.779, U_i (utilization) = 0.7166 i.e $U_i < U$ and is definitely schedulable.

Task1 Priority > Task2 Priority > Task3 Priority

Q.3 (b)

7	0	1	2	0	3	0	4	2	3	0	3	2	1	2	0	1	7	0	1
FIFO (Page Faults = 15)																			
7*	7*	7*	2	2	2	2*	4	4	4*	0	0	0	0	0*	0*	0*	7	7	7*
0	0	0	0*	0*	3	3	3*	2	2	2*	2*	1	1	1	1	1	1*	0	0
		1	1	1*	1	0	0	0*	3	3	3	3*	2	2	2	2	2	2*	1
LRU (Page Faults = 12)																			
7*	7*	7*	2	2	2*	2*	4	4	4*	0	0	0*	1	1	1*	1	1	1*	1
0	0	0	0*	0	0	0	0	0*	3	3	3	3*	3*	3*	0	0	0*	0	0
		1	1	1*	3	3	3*	2	2	2*	2*	2	2	2	2	2*	7	7	7*
OPTIMAL (Page Faults = 9)																			
7*	7*	7*	2	2	2	2	2	2	2	2	2	2	2	2*	2*	2*	7*	7*	7*
0	0	0	0	0	0	0*	4*	4*	4*	0*	0	0	0	0	0	0	0	0	0
		1	1*	3*	3	3	3	3	3	3	3	3	1*	1*	1	1	1	1	1

Q.4 (b)

FCFS:-

Current Cylinder	Next Cylinder	Total Cylinder Travel
53	98	45
98	183	85
183	37	146
37	122	85
122	14	108
14	124	110
124	65	59
65	67	2
		640

SCAN:-

Current Cylinder	Next Cylinder	Total Cylinder Travel
53	65	12
65	67	2
67	98	31
98	122	24
122	124	2
124	183	59
183	199	16
199	37	162
37	14	23
		331

SSTF:-

Current Cylinder	Next Cylinder	Total Cylinder Travel
53	65	12
65	67	2
67	37	30
37	14	23
14	98	84
98	122	24
122	124	2
124	183	59
		236

C-SCAN:-

Current Cylinder	Next Cylinder	Total Cylinder Travel
53	65	12
65	67	2
67	98	31
98	122	24
122	124	2
124	183	59
183	199	16
199	0	199
0	14	14
14	37	23
		382

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